

Application Development

App Design System

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3. Design System

Stack: Flutter with Material 3 (`useMaterial3: true`). Color, type, and spacing below are expressed as named tokens that map directly to `ColorScheme` , `TextTheme` , and spacing constants in `ThemeData` .

Step A: Color palette

Role	Your color (name or hex)	Used for
Primary	Coffee brown, #6F4E37 (seed color for Material 3)	Main buttons, active states, app bar
Secondary / accent	Tan, #D9B08C	Highlights, secondary actions, badges
Background	Warm off-white, #FBF6F2	Screen background
Surface	White, #FFFFFF	Cards, sheets, dialogs
Error	Material default red, #BA1A1A	Validation errors, delete actions
Text (on background)	Dark espresso, #2B1D14	Body text

Tokens: `color-primary` #6F4E37 · `color-secondary` #D9B08C · `color-background` #FBF6F2 · `color-surface` #FFFFFF · `color-error` #BA1A1A · `color-on-background` #2B1D14

Contrast check: `color-on-background` on `color-background` (#2B1D14 on #FBF6F2) ≈ 13.9:1, well above the 4.5:1 minimum. White text on `color-primary` (#FFFFFF on #6F4E37) ≈ 6.1:1, passes. Strength badge text on its tan-tinted fill (#8A5A2B on #F4E3D3) ≈ 4.6:1, passes at the 4.5:1 minimum for body-size text.

Step B: Type scale

Style	Size (sp)	Weight	Used for
Heading	22	Bold	Screen titles ("Calculator", "Notes")
Body	16	Regular	Input labels, list text, recipe details

Style	Size (sp)	Weight	Used for
Caption	12	Regular/light	Hints, timestamps, grind setting subtext

Tokens: `text-heading` · `text-body` · `text-caption`

Step C: Spacing rule

Base unit: 8px.

- Tight spacing (between related items): 8px
- Standard spacing (between sections): 16px
- Screen edge padding: 24px

Tokens: `space-tight` 8px · `space-standard` 16px · `space-edge` 24px

Step D: Reusable components

Sorted by level, atoms first, so each one is described once and reused everywhere it appears.

Atoms

Component	Appears on screens	Looks like	Props
Primary button	Home, Calculator, Timer, Notes	Filled rounded rectangle, primary color, white text	label, onPress
Outline button	Calculator, Timer	Rounded rectangle, outline only, primary-colored text	label, onPress
Text field	Calculator, Notes	Bordered rounded rectangle input	label, value, unit (optional)
Dropdown selector	Calculator (brew type), Roast + Grind (grinder)	Standard Material dropdown field	label, options, selected
Strength badge	Calculator, Notes	Small colored pill, text color matched to fill	label (Weak/Normal/Strong/Very Strong)

Molecules

Component	Appears on screens	Looks like	Contains
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Component	Appears on screens	Looks like	Contains
Ratio card	Calculator, Notes	Surface-colored rounded rectangle, 12px radius, light border	water grams, coffee grams, ratio, strength badge
Grind result card	Roast + Grind	Surface-colored rounded rectangle	roast toggle, grinder dropdown, grind setting badge
Timer display	Timer	Large centered digits, buttons below	current time, start/stop/reset controls
Recipe list tile	Notes, Home (recent recipes)	Row or small card, surface color, light border	brew method, ratio, date, strength badge

Organisms

Component	Appears on screens	Looks like	Contains
Calculator panel	Calculator, Roast + Grind	Stacked cards on mobile, side-by-side panels on desktop	Ratio card + Grind result card
Recipe list	Notes, Home	Vertical list or grid	Recipe list tile, repeated
Nav bar	Every screen	Bottom bar on mobile, sidebar on desktop	Calc, Notes, Timer, Settings links

Responsive breakpoints

- Below 600px (mobile): single column, one flow step per screen, bottom nav bar
- 600px and above (desktop/tablet landscape): 2-column panel grid, sidebar nav instead of bottom bar, Calculator/Roast+Grind/Timer/Notes shown together instead of stepped

Accessibility checklist

- All text/background pairs checked at 4.5:1 minimum (see contrast check above)
- Tap targets (buttons, dropdowns, nav items) sized at minimum 44x44px on mobile
- Nav icons (calculator, notes, clock, settings) paired with text labels, not icon-only
- Form fields (water/coffee grams, taste notes) use proper labels, not placeholder-only text, so screen readers announce them correctly
- Strength and grind-setting badges convey meaning through text label, not color alone
- Timer/stopwatch state changes (start/stop) confirmed visually, not just by color change

Coffee App — Design System

COLOR PALETTE

Primary
#6F4E37

Secondary
#D9B08C

Background
#FBF6F2

Surface
#FFFFFF

Error
#BA1A1A

Text
#2B1D14

TYPE SCALE

Calculator

Heading · 22sp · Bold

Water 200g, Coffee 33g

Body · 16sp · Regular

Saved 2 days ago

Caption · 12sp · Regular

SPACING RULE (BASE 8PX)

Tight — 8px

Standard — 16px

Screen edge — 24px

REUSABLE COMPONENTS

Ratio card

200g water / 33g coffee

Strong

Dropdown selector

Moka pot ▾

Timer display

02:45

Moka · Dark

Jul 9

Strong

Primary button

Coffee App

Recipes

Settings

RATIO CALCULATOR

Brew Type + ratio

Moka pot ▾

Water (g)

200

Coffee (g)

33

1:6.1–Strong

TIMER

02:45

Start

Stop

Reset

RATIO and Grind

Roast Type

Medium

Dark

Grinder

Comandante C40 ▾

18 clicks

Notes

Moka pot | 1:6.1 | Strong

Dark roast | 18 clicks | 02:45

Save recipe to Notes

MOBILE

HOME

Start Brewing

Recipes

Moka

Cold brew

French

RATIO CALCULATOR

Brew Type

Moka pot ▾

Water (g)

200

Coffee (g)

33

Ratio 1:6.1

Strong

Next: roast and grind setting

NOTES

Moka 1:6.1

Strong

Dark roast | 18 clicks | 02:45

Taste notes

nutty, slightly bitter

Temp (°C)

92

Save

RATIO and Grind

Roast Type

Medium

Dark

Grinder

Comandante C40 ▾

Grinder settings

18 clicks

Start Timer

TIMER

02:45

Start

Stop

Reset

Save & Continue